

Krishnaa Vadivel | GPU Software Engineer | High-Performance Computing | Scientific ML

📞 516-853-5663 • ✉ srikrishnaa.careers@gmail.com • in sri-krishnaa-ece-phd
Website | GitHub: krishnaa423 | LeetCode: krishnaa42342 | Docker Hub: krishnaa42342

Summary

GPU-focused computational scientist and software engineer targeting performance-oriented systems and HPC infrastructure roles. Builds accelerator-enabled applications in Python and C++ with CUDA, ROCm, MPI, and OpenMP, with experience in multi-GPU parallelization, collective communication, distributed debugging, and strong and weak scaling analysis on national-lab supercomputers.

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, SQL, C#

GPU / HPC: CUDA, ROCm, MPI, OpenMP, collective communication, multi-GPU parallelization, PETSc, SLEPc

Platforms: AMD MI250X, NVIDIA A100, Frontier, Perlmutter, Linux, CMake, Docker, Kubernetes

Debug / Perf: TotalView, Linaro DDT, distributed debugging, strong scaling, weak scaling, hardware-aware optimization

Machine Learning: PyTorch, PyTorch Geometric, automatic differentiation, equivariant graph neural networks, scikit-learn

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed distributed scientific software in Python and C++ for first-principles simulation workloads as part of an INCITE award using exascale resources, with MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier at OLCF and Perlmutter at LBNL.
- Parallelized matrix and vector workloads across multiple GPUs, using collective communication and distributed linear-algebra workflows to scale simulations across accelerator-backed clusters.
- Ported and optimized scientific-computing workloads for AMD MI250X and NVIDIA A100 GPU systems, translating many-body and excited-state physics methods into scalable accelerator-backed implementations.
- Built PyTorch and PyTorch Geometric models, including equivariant graph neural networks, that delivered roughly 100x speedups for parts of the workflow by combining model training with automatic differentiation to obtain additional quantities without separate expensive calculations.
- Ran strong- and weak-scaling studies to evaluate distributed performance and used TotalView, CUDA-aware MPI workflows, and Linaro DDT to debug multi-process and multi-GPU applications across heterogeneous compute environments.
- Packaged scientific software in Docker containers for reproducible deployment across local and HPC systems.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS research in 2024, 2025, and 2026, grounding GPU and HPC work in production research workflows.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built engineering software and supporting backend tooling for real-time remote tool testing, data collection, and sensor calibration workflows.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

FindLight

Photonics Technical Writer

2018–2019

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Accelerator-Enabled Exciton-Phonon Simulation: Developed distributed first-principles calculations with CUDA, ROCm, MPI, and OpenMP, including multi-GPU matrix and vector parallelization on leadership-class GPU clusters.

Scientific ML on GPU Systems: Developed PyTorch Geometric equivariant models that produced roughly 100x workflow speedups by pairing training with automatic differentiation to recover additional quantities efficiently.

CUDA Containers for HPC Software: Packaged scientific applications in CUDA-ready Docker environments for portable deployment across local and HPC systems.

Matrix-Multiply AI Accelerator: Built a Verilog-based matrix-multiply accelerator project for AI-oriented datapath design and hardware-aware compute fundamentals.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017